

1 PURPOSE

The Winston Solar Challenge is dedicated to the educational development of science students. On a broader scale, the race objectives are to stimulate interest in technical education, promote the use of renewable sources of energy, and raise public awareness of these issues.

2 ADMINISTRATION

- 2.1 **Application of Regulations** – The regulations will apply to the Winston Solar Challenge, hereinafter referred to as the “event.”
- 2.2 **Effective Date of Regulations** – The regulations become effective when issued in final form on January 1, 2004.
- 2.3 **Right to Revise Regulations** – The sponsor, as defined in Regulation 2.5 below, reserves the right to revise these regulations at any time by providing the participant written notice of the revisions in the form of Race Updates.
- 2.4 **Acceptance of Regulations** – All persons or groups selected to participate in the event are assumed to know the regulations. Their participation in the event will constitute acceptance.
- 2.5 **Sponsor** – The primary sponsor of the event is The Winston School, 5707 Royal Lane, Dallas, Texas 75229.
- 2.6 **Selection of Race Officials** – The sponsor will select race officials for this competition. These officials will be instructed in and be responsible for specific duties.
- 2.7 **Authority of Officials** – Winston Solar Challenge Officials will have the authority and responsibility to apply these regulations.

3 ENTRIES

- 3.1 **Selection of Entries** – Entries for the event are selected through a proposal system consisting of the evaluation of the application submitted by teams desiring to participate. The sponsor will select up to twenty proposals total from the classic and open divisions to participate in the event. The sponsor will also select up to 5 proposals from the demonstration division to participate in the event.
- 3.2 **Teams and Drivers** – Any number of people may act as the race team, however, during the event, it is suggested that a manageable core team of approximately twelve members be designated as “the race team.”
- 3.3 **Team Provisions** – During each Winston Solar Challenge, special meal opportunities are provided to the teams in the form of Open and Closing Banquets and one meal per day. The WSC will provide each team with identification tags and meal tickets for up to 12 team members and 2 chaperons per team. Larger teams must make reimbursement arrangements with the sponsor prior to arriving in Dallas. A special water facility will be provided to the teams at the race start, required stops, impound, and qualifiers. All other water requirements must be supplied by the teams.
- 3.4 **Makeup of Teams** – Teams, including drivers, will consist of high school students in grades 9-12. Twelfth grade includes the summer following the student's last year in high school. Team members must be from the same school district.

- 3.5 **Qualifiers** – Each team must successfully participate in a qualifier before they will be allowed to compete in the main event. Qualifiers will be held the day immediately preceding the race, as specifically announced in the Appendix. The qualifier will hereinafter be referred to as “scrutineering.”

Teams must also go through a pre-scrutineering process in June. Teams must make arrangements with the race staff for a visit to inspect their solar car. Teams will be required to present a 20-minute oral presentation to the race staff. Teams must also submit updated application and documents as specified in Section 3.6.

- 3.6 **Data Sheets** – Each team must submit data sheets no later than April 1st of the year in which they intend to race. The sheets must include all vehicle specifications including weight, dimensions, motor type, solar cell type, solar array power output, battery type and capacity, braking systems, and wheel type. All crew members, designated drivers, and faculty advisors must also be listed. The information from the data sheets will provide the basis for selection of event participants, and will appear in the Official Race Booklet.

Five additional documents must accompany the Data Sheets:

- (1) Complete, detailed drawing(s) showing the mechanical structure of the vehicle. This should be drawn with a drafting program such as AutoCAD for clarity and is not limited to one page. The drawing(s) must include crush zones, frame structure, and overall dimensions in three views (front, side, top). A final updated drawing must be submitted upon check in to the event if there has been any design changes since the first drawing was submitted.
- (2) A complete, detailed schematic showing the electrical layout of the vehicle. This schematic must include but does not need to be limited to all the wiring for the propulsion, solar, instrument, and battery systems. A final and current schematic must be submitted upon check in to the event.
- (3) A team photo, in color, at least 5 X 7 inches in size. The photo must clearly show the solar car and team members.
- (4) Manufacturer's data sheets for the propulsion system batteries, including, but not limited to voltage per battery, capacity per battery, weight of each battery, and proof that batteries are lead-acid.
- (5) Manufacturer's data sheets for the solar cells you intend to use, including, but not limited to the size of each solar cell, the efficiency rating, the number of watts per cell, and the list price of each solar cell.

Teams will forfeit \$100 of their refund [See Section 10.3] for failing to supply a complete, detailed drawing showing the mechanical structure of the vehicle, and an additional \$100 for failing to supply a complete, detailed schematic showing the electrical layout of the vehicle. Lack of detail or failure to supply a document may also result in teams forfeiting part of their refund.

Reasoning: Historically, teams have not submitted complete, final drawings of both mechanical and electrical layouts of their vehicle. Race Officials have concern that teams may arrive at the race with an unsafe vehicle, and not have enough time to make adequate changes to improve safety before the race begins. This would be heartbreaking for all concerned. If teams have to submit complete documentation earlier, then Race Officials will be in a better position to help and advise teams about potential problems. This should help reduce the chance of disqualification on the grounds of an unsafe vehicle.

- 3.7 **Changing Entry Data Sheets** – Entrants may change the data presented on their Data Sheets (3.6) up to the scheduled time of scrutineering, however, changes after June 1st may not appear in the Official Race Booklet.

- 3.8 **Vehicle Design and Construction** – It is the intent of the event that the solar cars be designed and constructed by high school students. The major components of the vehicle (frame, body, solar array, battery compartment, motor mount, drive system, electrical system, suspension, running gear, and lighting system) must be designed and constructed completely by the students. Off the shelf components (e.g., wheel hubs, brake calipers, power trackers, motors, motor controllers, suspension/steering components, wheels, etc.) may be used as long as the selection of these components was made exclusively by the students. Teams can receive contributions of off-the-shelf components from other solar teams.

The students may seek the advice of engineers or other design consultants. Parts may be fabricated by specialty shops (for welding, machining, etc.) as long as the students specify what the specialty shop is to do, through drawings or specification sheets prepared by the students. However, it is the spirit of the event to learn how to build a solar car, and it is recommended that students perform whatever tasks they can to construct the solar car on their own. During scrutineering, judges will strenuously question the students to determine compliance with this rule.

4 OFFICIAL RACE COURSE

The Official Race Course on even-numbered years will be the Texas Motor Speedway. The Official Race Course for odd-numbered years will be announced in the Appendix to these Rules.

5 SOLAR CAR REGULATIONS

- 5.1 **Solar Car Dimensions** – All solar cars entered will have the following maximum dimensions: length = 6 meters; height = 1.6 meters; width = 2 meters. Minimum height is 1 meter. When turning corners, wheels and wheel fastenings may exceed these dimensions. A height exception will only be granted to antennas.

In 2004, solar cars will be allowed to conform to the ISF 5000 (formerly NIS) standard, which will have the following maximum dimensions: length = 5 meters, height = 1.6 meters, width = 1.8 meters. The minimum height for all solar cars will remain 1 meter. In 2005, all newly-built solar cars must conform to this standard. Solar cars entered in previous events may continue to adhere to traditional dimensions until the 2007 race year.

- 5.2 **Structure** – Safety is the Sponsor's primary concern with regard to the structural development and fabrication of the solar cars.

Insufficient regard for structural safety will result in disqualification from the event during scrutineering. In specific, judges will require teams to show a specific *crush zone* designed to help protect the driver in the event of collision. Teams not being able to demonstrate a specific crush zone will receive a significant penalty and risk possible disqualification from the race.

Crush zone is defined as an area of the car designed to collapse in an effort to absorb some of the impact from a collision without impacting the driver space. A crush zone must be considered for frontal, side, and rear collisions. Therefore, the driver, when seated, must have a minimum of 15 cm of horizontal distance from his or her shoulders, hips, and feet to the inside of the most exterior structural frame member.

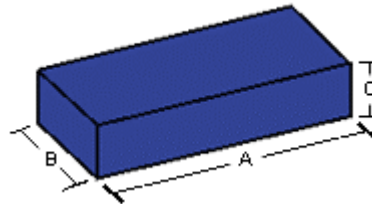
- 5.2.1 **Stability** – To ensure stability under various road conditions, the vehicle must employ either a three or four wheel design. If a three wheel design is used, the WSC strongly suggests that the single wheel should be in the rear of the vehicle, with two wheels in front. During scrutineering, cars must demonstrate the stability of the vehicle. Either the front or rear wheel(s) or both may be powered, and either the front or rear wheel(s) may be used to steer the vehicle.

5.2.2 **Body of the Vehicle** – Although windshields are required, solar cars need not have a body or covering unless the race judges determine that the driving of that vehicle would be unsafe and/or create an unreasonable risk of harm to the driver.

5.2.3 **Roll Cage** – All solar cars must be equipped with a roll cage that encompasses the entire driver. The roll cage shall be a fixed, integral part of the solar car structure. The protection provided for the driver in a collision must be documented in the team's mechanical drawings. In addition to providing collision and rollover protection, the roll cage must be designed so as to deflect body/array panels of the car away from the driver in the event of an accident. There must be at least 5 cm of clearance in all directions between the roll cage and the driver seated in the normal driving position. The roll cage tubing must be made from carbon steel and have a minimum outside diameter of 5 cm and minimum wall thickness of 1.5 mm. If aluminum is used, the roll cage tubing must have a minimum outside diameter of 5 cm and a minimum wall thickness of 3.2 mm. Alternate materials that afford equivalent protection for the driver are permitted, provided they are fully documented in the team's mechanical drawings.

5.3 **Power** – Global solar radiation received by the car directly from the sun is the only power source that will be used for propulsion.

5.3.1 **Solar Cell and Array Size Limitations** – Only terrestrial-grade solar cells will be allowed, except for the provisions set out in Section 12. The solar collection panel may be constructed in any way, provided that it lies, at all times, entirely within a single right rectangle parallelepiped of dimensions $A \times B \times C$, where A is less than length 4.44m, B is less than width 2.00m, and C is less than height 1.60m.



In addition, the linear area of the solar array must fit within an 8-m^2 area, although a single rectangular region not occupied by solar array components may be excluded from the linear area calculation. When in motion, the parallelepiped must be oriented such that A , B , and C are length (measured horizontally), width (measured horizontally), and height respectively.

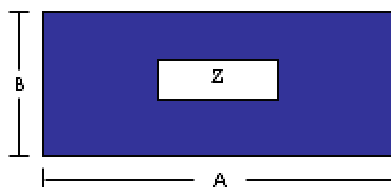
When stationary, the right rectangle parallelepiped may have any orientation. The solar collection panel must remain mounted on the outside of the vehicle, except as provided in Section 5.19; the panel cannot be dismantled and carried inside the vehicle.

The panel cannot be reconfigured during the course of the race or when stationary, without specific written permission of the Sponsor. No reflector-type device can be used to enhance the collection capacity of the panel.

All leads extending from the solar array must be properly protected to eliminate accidental shock hazards resulting from participants or bystanders coming in contact with these leads.

In 2004, solar cars will be allowed to conform to the ISF 5000 (formerly NIS) standard, which allows the solar collection panel to be constructed in any way, provided that it lies within the maximum dimensions of the solar car. In 2005, all newly-built solar cars must conform to this standard. Solar cars entered in previous events may continue to adhere to traditional dimensions until the 2007 race year.

- 5.3.2 **Measuring the Array** – The area of the solar array will be calculated by measuring the length and width from the outside edges of the active part of the solar cells. Spaces or gaps between the cells will be included in the total area. The area within the single rectangular region described in Rule 5.3.1 is not included in determining the size of the array. Regardless of the size of the canopy, the entire solar array must fit within the single rectangle parallelepiped in Rule 5.3.1.



For example, in the diagram above, the area of the solar array equals $(A \times B) - (\text{area of } Z)$, where Z is the rectangular region within the linear area of the solar array where there are no solar array components.

In 2004, solar cars will be allowed to conform to the ISF 5000 (formerly NIS) standard, which allows the solar collection panel to be constructed in any way, provided that it lies within the maximum dimensions of the solar car. In 2005, all newly-built solar cars must conform to this standard. Solar cars entered in previous events may continue to adhere to traditional methods for measuring the array until the 2007 race year.

- 5.3.3 **Supplemental Batteries** – Any batteries that are used to power solar car accessories, such as, but not limited to, radios, electronic speedometers, cameras, memory devices, computers (only if used for data gathering and not to control the vehicle systems), and telemetry transmission must be totally isolated from the propulsion system of the solar car. At scrutineering, these batteries, and the devices they are in, will be checked to ensure that no possibility exists to convert this power to propulsion for the solar car. Also supplemental batteries will be allowed to power the following equipment: (1) lights; (2) horn; and (3) battery box air circulation fans, so long as the driver is aggressively notified when the supplemental battery pack is running low. The driver must have sufficient warning to safely pull off the road to replace the supplemental battery. Teams must be able to demonstrate their warning system to the satisfaction of the judges during scrutineering. There is no weight limitation placed on the supplemental battery system. Supplemental batteries do not have to be lead acid. The supplemental batteries must be enclosed in a battery box that is separate from the main battery pack.
- 5.4 **Motors/Gear Ratios/Tires** – There are no regulations limiting the selection of motors, tires, or gears on solar cars. It has been noted, however, that tires undergo greater wear on their sides during the closed-track events as compared to the cross-country events. This unusual wear can be attributed to the slant of the closed track.
- 5.5 **Storage of Solar Radiation** – All solar cars in the event will be allowed to store solar-generated energy.
- 5.5.1 **Battery Type** – In the interest of cost and standardization, only rechargeable, commercial production, lead-acid batteries are permitted. The total weight of the batteries (alone) can not exceed 275 pounds. There is no limit applied to system voltage, number of cells, or modules.
- 5.5.1.1 **Battery Weight Penalty** – Teams that exceed the weight limitation will be penalized 2.5 miles per pound over the 275 pound limit.

- 5.5.2 **Battery System** – Batteries must be fully contained in enclosures that can be sealed by race officials (thus sealing the batteries). A hasp latch for the battery enclosure will assist race officials in complying with this section of the rules. Battery enclosures will be equipped with a forced air ventilation system, which must operate whenever the battery system is electrically connected to the solar car. Such ventilation system must exhaust to the exterior of the solar car. All electrical cables must be properly sized to expected system currents. Teams that choose to power their venting fans from a supplemental battery must comply with the requirements set out within section 5.3.3.

A battery enclosure is defined as a rigid box-like structure that provides protection and stability to the battery system in the event of collision. Battery bag devices are not appropriate. The battery enclosure must be secured to the chassis of the vehicle.

- 5.5.3 **Main Fuse** – A separate fuse (not a circuit breaker) will be placed in series with the main battery. The fuse's rating will not exceed 200% of the maximum expected current draw. All low voltage taps from the main battery will be separately fused. All fuses must be placed first in series with the battery starting at the positive connection.

- 5.5.4 **Array Disconnect** – The solar charging system will be equipped with a manually operated high current switch to quickly disconnect the solar array from the battery pack. The array disconnect must directly interrupt the current from the array to the battery and must be capable of carrying and interrupting the peak array current. This device will not operate a separate contact, relay, or solenoid switch.

The switch must be clearly labeled in 10mm-high letters as the "Array Disconnect" and be plainly marked (10mm-high letters) with "ON" and "OFF" positions.

The switch must physically be located as near to the battery pack as practical and be operable from outside the solar car. This means that the canopy or motor compartment does not have to be opened first to activate the disconnect.

When both the array disconnect and the motor disconnect are thrown, the solar array, battery pack, and motor must be disconnected from each other.

- 5.5.5 **Motor Disconnect** – Solar cars must have a "kill" switch wired to disconnect all power to the motor. The motor disconnect must directly interrupt the current through the motor and must be capable of carrying and interrupting the full load motor current.

The switch must be clearly marked in 10mm-high letters as the "Motor Disconnect," and be within easy reach from both the driver's position and from outside the solar car. The switch must be plainly marked (10mm-high letters) with "ON" and "OFF" positions.

The motor disconnect must be separate from the array disconnect. This device will not operate a separate contact, relay, or solenoid switch.

In the case where a brushless DC motor is used, the motor disconnect switch may be placed in the main DC current loop supplying power to the motor controller, or alternatively, at least two of the three main leads between the controller and the motor must be interrupted by the motor kill switch.

Two switches wired directly in series may be used so that one of the switches is within easy reach from the driver's position and the other switch is within easy reach from the outside of the solar car. If two switches are used, each switch must be able to interrupt full load current.

When both the array disconnect and the motor disconnect are thrown, the solar array, battery pack, and motor must be disconnected from each other.

- 5.5.6 **Assistance Devices** – Any device used to assist the start, stop, or powering of a solar car must be carried on the solar car and must be a permanent part of the electrical system. Once the race has started, teams will not be allowed to plug any devices into the solar car except telemetry equipment and multimeters.
- 5.6 **Seating Position** – All seating must position the occupant's head higher than his or her feet. No head-first positioning will be allowed for the driver.
- 5.7 **Visibility** – In the normal driving position, each driver's eyes must be a minimum of 70cm above the ground.
 - 5.7.1 **Forward Visibility** – From the position identified in section 5.6, above, all drivers must be able to see, without artificial assistance: (1) a point on the ground 8 meters in front of the solar car; and (2) a minimum of 10 degrees above the horizon on level ground.
 - 5.7.2 **Side Visibility** – From the position identified in section 5.6, above, all drivers must be able to see, without artificial assistance, 90 degrees to either side at all times. This must be essentially unobstructed by the solar car structure. The intent is for the driver to see oncoming traffic, left and right.
 - 5.7.3 **Rear Visibility** – All solar cars must be equipped with a rear vision system that will allow the driver to see a vehicle 15 meters directly behind the solar car, as well as see vehicles 45 degrees in each direction approaching from the rear. If an electronic rear vision system is used, it must operate whenever the solar car is operating under its own power.
- 5.8 **Braking** – The intent of the Sponsor is to require that solar cars be capable of braking and coming to a safe stop based upon road and traffic conditions. To this end, solar cars will be equipped with two separate, independent braking systems. This means that there must be two separate hand or foot pedals (one for each braking system) and that each is connected to a different actuating device. Regenerative brakes are not considered a separate, independent braking system for consideration in this rule.

During scrutineering, solar cars will be required to demonstrate their braking capability at both 20 mph and at the vehicle's maximum speed.

 - 5.8.1 **Low Speed** – At 20 mph, the driver must be able to bring the vehicle to a safe, controlled stop in no more than 2.5 seconds.
 - 5.8.2 **High Speed** – At the vehicle's maximum speed, the driver must be able to bring the vehicle to a safe, controlled stop without drifting out of the appropriate lane of travel, and within a reasonable time frame based on that vehicle's maximum speed.
- 5.9 **Turning Radius** – Solar car wheels must be able to make a U-turn in a 15-meter wide lane.
- 5.10 **Warning Systems**
 - 5.10.1 **Lighting** – Solar cars must have stoplights, front and rear turn indicators, and hazard lights visible from 30 meters away. The geometric visibility of each light should be 45 degrees from center and 15 degrees up and down. Teams will receive a "moving violation" for each incidence of non-working lights. Once the team is notified by a race official that they have a non-working light, they must safely remove themselves from the race and make the necessary repairs.

- 5.10.2 **Audible Warning** – Drivers must be able to give audible warning to pedestrians and other vehicles using a horn rated at greater than 92 decibels. If the horn is located inside the passenger compartment, it must be properly secured prevent injury during a collision.

In closed-track races, the horn must be sounded when a solar car attempts to overtake and pass another vehicle.

- 5.11 **Driver Safety** – Teams must take whatever steps necessary to protect the safety and well-being of the drivers. Driver safety measures include, but are not limited to:

- 5.11.1 **Safety Belts** – Solar cars must be equipped with a minimum of a three-point lap and shoulder belt. The use of safety belts is mandatory at any time the vehicle is moving. The belts must be attached securely to a structural component or main frame member of the solar car. The harness must be attached with bolts at least grade 8 and 3/8" in diameter.

- 5.11.2 **Impact Protection** – The driver compartment must be equipped with structural components that help protect the driver from vehicle roll-over, or from front or side impacts, in compliance with Rule 5.2 (above).

- 5.11.3 **Protection from Vehicle Components** – All equipment housed within the vehicle must be adequately secured to the main frame to prevent the shifting of that equipment during an impact. Special emphasis is placed on securing the batteries due to their weight and potential for environmental damage.

- 5.11.4 **Windshield** – All solar cars will utilize windshields at all times to protect the driver from road hazards.

- 5.11.5 **Cockpit Egress** – The driver must be able to demonstrate that unassisted egress can be achieved in less than 15 seconds in the event of an emergency. The "exit process" will be carefully checked by race officials during scrutineering. Other team members can not assist the driver during this required "exit process."

- 5.11.6 **Fire Extinguishers** – Cockpits must be equipped with a fire extinguisher which can be easily reached by the driver in the event of an emergency. Chase vehicles must also be equipped with a fire extinguisher.

- 5.11.7 **Liquid Requirement** – Cockpits must be equipped with a plastic liquid container filled with water, which can be easily reached by the driver.

- 5.11.8 **Belly Pan Requirement** – The cockpit must be equipped with a full belly pan to isolate the driver from the road.

- 5.11.9 **Ultimate Driver Safety** – Each team must recognize that the ultimate responsibility for driver safety rests with the individual solar teams and not with the Winston Solar Challenge, or its officials.

- 5.11.10 **Air Circulation Requirement** – Forced air ventilation must be provided for the solar car's driver. This is particularly important for the health and safety of the driver while the vehicle is powered but not in motion. Power for this system can come from a supplemental battery. This requirement applies whether or not the driver's compartment is enclosed or open.

- 5.12 **Throttle** – Accelerator mechanisms on solar cars must be free moving and when released, must return to the zero current position. If the solar car is equipped with cruise control, it must be designed with an automatic shut-off when the brake is activated.

5.13 **Covers and Shields** – The solar car's revolving parts must be suitably covered to prevent accidental contact. All steering linkage must be shielded from the contact of the driver. If a flywheel is used, it must be covered by a National Hot Rod Association-approved shield.

5.14 **Electrical Shock Hazards** – All conductors must be properly insulated. All conductors operating at greater than 36 volts must be marked with "High Voltage" warning signs. All un-encapsulated solar cells and panels must be marked with "High Voltage" warning signs.

All leads extending from the solar array must be properly protected to eliminate accidental shock hazards resulting from participants or bystanders coming in contact with these leads.

Encapsulation is defined as any covering designed to protect the solar cells and prevent shock hazards. This includes, but is not limited to, glass, plastic, resins, and acrylic coverings.

Winston Solar Officials strongly suggest the following steps to improve safety:

- (1) When working with the battery box, be sure that you only use one hand, and that the other hand is kept behind your back. In the event of an electrical shock, the charge will not pass through the entire body.
- (2) When working with the battery box, be sure to use rubberized tools to insulate against the possibility of electrical shocks.
- (3) Students working in the battery box require continuous supervision by teacher/sponsors.
- (4) When working with the battery box, be sure to wear eye protection (a full face shield is best) to protect against battery explosions caused by sparks or arcing.

5.15 **Radios** – Solar cars must be equipped with a two-way radio to allow communication with the chase vehicle. Teams must demonstrate during scrutineering that the transmission from the solar car can be received at a distance of at least 0.5 miles. If a radio system "goes down" during a racing event, drivers must immediately withdraw from the race until repairs are made. The clock continues to run during these repairs.

Judges will make every effort to ensure that a team's radio communications will remain private, unless there is a safety issue involved.

5.17 **Fasteners** – All fasteners associated with the vehicle's suspension, steering, brakes, seat belts, battery chassis, and drive train must be equipped with locking nuts, double nuts, or nuts secured with safety wire or cotter pins. *Loctite* may be used in areas of difficult accessibility.

5.18 **Graphics**

5.18.1 **Inappropriate Graphics** – The race organizer reserves the right to disapprove any graphic it deems offensive.

5.18.2 **Sponsored Graphics** – Teams must reserve a space 4 x 8 inches on each side of the solar car for sponsored graphics.

5.19 **Umbilical Cord** – To position a solar car's solar array in a favorable position for charging, an umbilical cord may be used. This cord must be no longer than 3 meters, and must be carried on the solar car.

The purpose of the umbilical cord is to provide the racing participant the means to keep the array electrically connected to the vehicle, though physically removed from the vehicle to receive more favorable sunlight.

6 NATURE OF THE EVENT

In even-numbered years, the event will consist of a 3-day closed track race. Teams will cover the same course each day with the same specific start and finish line at the Texas Motor Speedway. The team that completes the most miles during the 3 days of the race will be declared the winner. The details of even-numbered year events will be spelled out in *Rules Updates* for that specific year's event.

On odd-numbered years, the event will consist of a cross-country race. Teams will race three days, rest one day, and race an additional 3-4 days. The team with the most miles over the course of the race will be declared the winner. The details of odd-numbered year events will be spelled out in *Rules Updates* for that specific year's event.

7 DATES AND LOCATIONS

7.1 **Race Events** – Teams will be required to take part in all special race events and activities as set out in the *Rules Updates*. Additional events may be added to this list. Teams will be notified at Team Check-In which precedes the Opening Banquet.

7.2 **Items Provided by the Winston Solar Challenge**

- Comprehensive Race Booklet
- Guide to Hotel/Motel/Campsite Accommodations
- Emergency Medical Personnel
- Judges Accompanying Each Team
- On-road police protection (where appropriate)
- One meal per day from the Opening Banquet to the Closing Banquet (For up to 12 team members and 2 Advisors)
- Special Texas License Plate (For cross-country events only)
- *Caution: Solar Car Ahead* Sign
- Drinking Water (At Scrutineering, Start & Finish Lines, Media Stops, and Vehicle Impound)
- Security for Solar Cars during nightly impounds

7.3 **Items Provided by WSC Entrants** – All items not specifically enumerated in Section 7.2 (above) will be provided by entrants, unless set out in *Rules Updates*. Teams must remember to provide driver's licenses for designated Team Drivers, as well as Proof of Vehicle Liability Insurance.

7.4 **Trailer Provisions** – The cross-country race course contains specific trailering provisions. Please refer to the *Rules Updates* for that specific race. These provisions may change from one cross-country race to the next, depending on road conditions.

8 SCHEDULED TIMES

Central Daylight Time is the "official" time for all race events. When a cross-country race moves into another time zone, the time within the zone when the race day started will be considered the "official time" for all race events conducted in that new time zone on that race day.

8.1 **Racing** – Each day of racing will begin at 9:00 AM. Each team will be released from an official Starting Line at 3-minute intervals and all racing will stop 8 hours after the team is released.

8.2 **Charging** – Teams may charge their solar car batteries during daylight hours up to the time for impounding the vehicles. Charging of all batteries, including the main battery bank and supplemental batteries, must be done in the presence of a race official. The main battery bank can only be charged by direct solar radiation.

- 8.3 **Impound** – All solar cars will be impounded under event security at 9:00 PM each day. Teams may start removing their solar cars from the Impound after 6:00 AM. Once the race has begun, no team will be allowed to remove their vehicle from the Impound without the escort of a race official. The car must remain under the direct supervision of a race official throughout the day until it is again submitted for impound. Teams violating this rule may be disqualified from the race.

9 SCRUTINEERING

- 9.1 **Time** – Solar cars will be inspected to verify compliance with the regulations at 8:00 AM on the day preceding the race. Vehicles not ready to begin scrutineering at this time will be penalized up to 5 miles.
- 9.2 **Inspected Solar Cars** – Solar Cars that pass scrutineering will be allowed to participate in the race. Vehicles that do not pass scrutineering must correct the deficiency in order to be considered for participation in the event. After repairs are made, the team has the responsibility to notify race officials that their vehicle is again ready for scrutineering. Vehicles will continue to be inspected until 12:30 PM on the day preceding the race. Scrutineering will officially end at this time unless an extension is specifically announced by race officials.
- 9.3 **Safety** – Each team is responsible for the road-worthiness of its solar car. Passing the scrutineering process does not relieve the team of its responsibility, nor impose any liability on the race organizers. Solar Cars must be maintained in a safe, road-worthy condition at all times.
- 9.4 **Liability** – Teams will be required to sign a Release of Liability Form prior to admission to the event. Proof of Vehicle Liability Insurance will also be required at the time of scrutineering must be submitted no later than June 15th.
- 9.5 **Spare Batteries** – Any spare batteries a team wishes to utilize for possible replacement during the event must be inspected and tagged during scrutineering.
- 9.6 **Braking** – Solar cars will be required to meet the braking guidelines set out in Section 5.8. Failure to meet these guidelines will require the team to make the necessary changes to bring their vehicle into compliance, and then to be re-tested. Any solar car that does not pass the brake test will not be allowed to enter the race.
- 9.7 **Handling** – Solar cars will be required to run through a slalom course to ensure proper handling when lateral forces are exerted on the frame and wheel assemblies.
- 9.8 **Safety Meeting** – Team drivers and advisors will be required to attend a safety meeting scheduled during the scrutineering process.
- 9.9 **Safety Equipment** – Each team must demonstrate during scrutineering that it is prepared to handle emergencies. This includes the designation of a *team safety coordinator*. The following equipment must be easily accessible to the team: fire extinguishers (both on the solar car and the chase vehicles), traffic cones or warning triangles, flags, safety vests, and jack stands. Jack stands must be used whenever the solar car is off the ground for maintenance.
- 9.10 **Post-Race Inspection** – At the conclusion of the event, the race officials will conduct a post-race inspection of the first place team in each division. In addition, race officials will have the discretion to conduct post-race inspections for all other teams. The post-race inspection will confirm that the car complies with event regulations and that the components in the car agree with the final Data Sheets (Section 3.6) submitted to the event. If a car fails any portion of the post-race inspection, the team will be penalized based on the provisions set out in section 27.

10 REGISTRATION

- 10.1 **Purpose** – The purpose of registration is to identify participants and vehicles taking part in the event.
- 10.2 **Registration Deadline** – Teams must file registration papers no later than April 1st of each race year.
- 10.3 **Fees** – The registration fee is \$500. A check for \$400 will be refunded to the team upon arrival and qualification at the Texas Motor Speedway, except in the following instances:
 - (a) Teams registering after April 1st will not be entitled to the refund.
 - (b) Teams will forfeit \$100 of their refund for failing to supply a complete, final drawing showing the mechanical structure of the vehicle, and an additional \$100 for failing to supply a complete, final schematic showing the electrical layout.
- 10.4 **License Plates** – No license plates are required on the closed-track events. The WSC will supply a 30-day special Texas License Plate for cross-country events.
- 10.5 **Participants Registered** – Any person taking part in the event must be registered, and must wear an official WSC Identification Badge around their neck at all times. This includes media, sponsors, officials, guests, and teams. The Identification Badge must be visible at all times.
- 10.6 **Driver Restrictions** – Solar car team drivers must have a valid driver's license. All drivers will be specifically identified at registration. Only registered drivers will be allowed to drive during the event. On each day of the race, solar car drivers must report the name (and function) of any drug that they are using. The report should be made during the drivers' meeting to the Assistant Race Director.
- 10.7 **Driver Weight** – No ballast provisions will be imposed on any team.
- 10.8 **Liability Insurance** – Teams must submit proof of liability insurance on their solar car no later than June 15th of the year in which they intend to race. Teams not complying with this rule will receive an automatic penalty during Scrutineering.

11 STARTING ORDER

The starting order for the first official day will be determined by a drawing at the Opening Banquet. On all other days, the starting order will be determined by the solar cars' previous day mileage.

The solar cars will be released from an official Starting Line at 3-minute intervals. If a solar car is not ready for its assigned starting slot, all following cars will move up one slot, and the detained car will become the last on the grid.

12 LEVELS OF PARTICIPATION

- (1) Race participants must choose to participate in one of three divisions of racing:

CLASSIC DIVISION – Classic Division retains all the rules and regulations set out in this race booklet. No hub motors can be used. Solar cells must be rated below 16% efficiency.

OPEN DIVISION – Open Division provides teams the option to implement new, more costly technologies within their solar car design or vary the dimensions of their solar car. If a team uses a hub motor, or utilizes terrestrial or space-grade solar cells that have a rated efficiency of 16% or above, they must enter the Open Division. The size limitations of the solar array remain the same, however, individual solar cells may be less than 100 square centimeters. The list price for all bare solar cells must be less than \$10/watt; teams may pay extra for cutting, tabbing, or lamination of cells. The limitations on type and weight of batteries remain the same.

DEMONSTRATION DIVISION – Demonstration Division provides a way for Winston Solar Challenge alumni to participate in this race. Teams entering this division must be composed of high school graduates, and the majority of team members must have previously participated in a Winston Solar Challenge event. Vehicles entered into the Demonstration Division are limited to the same rules and regulations as the Classic Division.

- (2) Trophies and Awards will be granted in both the Classic and Open Divisions.
- (3) The judges will select one vehicle to receive the World Solar Challenge Award, based upon which team would best represent the Winston Solar Challenge at the next World Solar Challenge. Factors to be considered by the judges include: (a) speed; (b) efficiency; (c) endurance ability; (d) results of the overall race; (e) team dedication to the spirit of solar racing; and (f) good sportsmanship.

13 SCORING – Official times and distances are the responsibility of race officials. The details for the scoring of each race will be set out in that event's specific *Rules Updates*. Daily race results will be posted on the race trailer prior to the start of the next day's competition.

14 DRIVER MEETING – Drivers and Team Sponsors must attend a briefing at 8:00 AM each race day. A major penalty will be imposed on any team not attending this important session.

15 OVERNIGHT

- 15.1 **Impound** – The impound will be a building, tent, or protected parking lot where every solar car will be secured under the direction and supervision of race officials. There will be no vehicle maintenance allowed on the solar cars while in the Impound. Solar cars transported to the Impound and arriving after 9:00 PM will be impounded immediately upon arrival.
- 15.2 **Garage** – The garage will be an area for charging and maintaining vehicles before and after the impound. No other vehicles will be permitted in the garage area.
- 15.3 **Support Vehicle Parking** – All support vehicles must be parked in this area. Maintenance or repair of solar cars requiring close proximity to their support vehicles must be done in the support vehicle parking area.
- 15.4 **Headquarters** – Event Headquarters will be located near the Impound and Garage area. All official announcements and postings will be at the Headquarters.
- 15.5 **Accommodations** – Teams are responsible for securing their own accommodations. In Dallas, the official WSC Hotel is the Dallas Parkway Hilton [972.661.3600]. On cross-country races, a special list of event hotels will be mailed to teams prior to the event so that timely reservations can be secured. Special team prices are available at all event hotels.

16 MANDATORY STOPS – Teams are required to make specific stops during the race day. Solar charging, vehicle maintenance, team needs, and driver rotation are allowed during these stops. Teams are “off the clock” during the time that they are within an official “stop” area.

Specified Stops – At Team Check-in, teams will receive a detailed list of stops. These include media opportunities, organized lunch breaks, vehicle inspections, or required trailering. All specified stops are “off the clock.”

Un-Specified Stops – On occasion, team members may need to make an unspecified hygiene stop. If this occurs, the team should plan ahead so that the solar car can continue on the race. Chase cars can be swapped so that team members can seek necessary relief, so long as adequate provision is made for the race official to continue on with the solar car. Any unspecified stop is “on the clock.”

17 CHECKPOINTS – Checkpoints will be established along the course. Failure to pass a checkpoint may result in race disqualification.

18 MAINTENANCE – Vehicles can be repaired and maintained at any time during the day except when public or media events make it inappropriate, or during the impound. Any component of the solar car may be changed or repaired with the following restrictions.

18.1 Batteries – Cells, modules, or battery packs may not be replaced for any reason other than malfunction or accident. Decisions to exchange all or part of a battery system must be communicated first to a race official who will record the exchange and penalty. The judge must observe the breaking of any battery seal, date, and initial the broken seal. The chief scrutineer has the authority to impose further penalties if a team, in effecting a battery replacement, has gained an unfair advantage over the other teams or has otherwise departed from the spirit of the competition.

$\frac{\text{Number modules replaced}}{\text{Total number of modules}} \times 100 \text{ miles} = \text{Penalty}$

19 ACCIDENTS – All accidents must be reported to a race official immediately. Failure to do so may result in expulsion from the event. In the case of an accident involving personal injury and/or property damage, notification of the appropriate emergency personnel will have priority.

19.1 Re-Inspection – If a solar car is involved in an accident, it must be re-inspected by race officials. If the safety of any solar car is in question during the race or during the scrutineering process, a race official may require re-inspection prior to that vehicle's resuming the race or resuming the scrutineering process. No time will be charged to that vehicle's team during the safety inspection.

20 WITHDRAWALS – Any team that has agreed to participate must fulfill its obligation unless specifically excused by the race organizer. Any team wishing to withdraw must notify the organizer in writing.

21 PUSHING/TRAILERING – Solar cars may be pushed under the following conditions: (1) Into and out of the required Impound; (2) After the team is “off the clock” at Mandatory Stops; and (3) During an “on-the-clock” situation when a race official authorizes “pushing” to protect the safety of the team. Trailering will be allowed as specifically set out in that event's *Race Updates*.

22 SUPPORT VEHICLES – Teams must have all their support vehicles registered and properly marked at the time of scrutineering.

- 22.1 **Cross-Country Support Vehicles** – Teams will be allowed two support vehicles on the course with the solar car: a lead vehicle (driven with lights on) directly in-front of the solar car, and a chase vehicle directly behind the solar car. All support vehicles on the race course must have roof-mounted flashing amber lights. The chase vehicle must also carry a rear-mounted sign (supplied by the race organizer) that reads: “Caution! Solar Car Ahead.” If the team has a lead vehicle, it must be no further than 10 vehicle lengths ahead of the solar car, nor closer than 4 vehicle lengths ahead of the solar car to prevent “drafting.” The lead vehicle is required whenever the solar car is driven on the highway. The chase vehicle must remain within 3-5 vehicle lengths from the solar car at all times, including media stops, rest stops, trailering, repairs, etc. It is suggested that the lead vehicle pull the trailer that stores the solar car. Any other support vehicles must keep a distance away from the lead and chase vehicle to allow traffic to pass.

Teams must provide space in the chase vehicle for a race judge and race telemetry equipment.

- 22.2 **Closed-Track Event** – Teams will be allowed one support vehicle in the event that the solar car stops on the track. This support vehicle must have a roof-mounted flashing amber light. The support vehicle will only carry enough equipment to facilitate the solar car's return to the pit area. No significant repairs will be made on the track itself. In the event that the car must be trailered, teams are required to first get the permission of the Chief Race Official so that removal of the vehicle from the track can be carried out without undue hazard to the other racing solar cars.

- 22.3 **Accompanying** – At any time the solar car is moving on the open road, it must be accompanied by a designated chase vehicle (with the flashing amber lights and rear-mounted sign). Failure to follow this safety regulation may subject the team to a major penalty.

- 23 **OVERTAKING** – Situations will arise when slower solar cars (and their accompanying support vehicles) will need to pull over when being overtaken by a faster-moving solar car. Failure to pull over, or recklessly pulling out into the path of a passing solar car, will result in a major penalty. In the event that one team is overtaken by another team, the faster team must signal their intention to pass by flashing the lead vehicle's headlights. The slower team must then give way by slowing down or pulling over to allow the faster team to pass. Safety is the primary consideration here.

- 24 **DRAFTING** – Drafting by a solar car is prohibited. Solar cars must be no closer than 4 vehicle lengths behind the car in front of them; solar cars must be no closer than 3 vehicle lengths to their chase vehicle behind them.

- 25 **HELMETS** – Helmets are not mandatory for solar car drivers during the race, but are highly recommended. The design of the vehicle must provide sufficient protection to the driver's head in the event of a rollover, or collision.

26 JUDGING

- 26.1 **Team Action** – A significant technique to help teams not incur penalties is for the team to designate a liaison to inform the race official about what the team is going to do before the team does it! This includes how the team will deal with emergencies, intentions to trailer, plans to switch-out drivers, plans to make unspecified stops, etc.

26.2 Judges

- (1) These rules are in addition to, and do not contradict any existing rule.
- (2) Each day, judges will be assigned, on a rotating basis, to solar teams. Teams must provide a seat in the chase vehicle with an unobstructed view of the solar car for the Team Judge. A judge (meaning the adult judge or the intern/judge) must remain in close proximity to the solar car at all times.

- (3) Judges will be required to enforce all rules set out in the Rules or Appendices.
- (4) Judges will be rotated daily.
- (5) Judges are prohibited from supporting or giving any tactical advice to a team during the race. Any judge violating this rule will be immediately removed from the race.
- (6) Judges will make every effort to ensure that a team's radio communications will remain private, unless there is a safety issue involved.
- (7) Judges must report their comments and penalties to the headquarters judge at the end of each race day. In addition, judges must report any favoritism shown to any team. A history of "errors" or poor record-keeping may cause a judge to be removed.
- (8) Any team protest involving a judge must be filed no later than 6:00 PM on the day on which the alleged infraction occurred.

26.3 A Team's Duty to their Judge on Cross-Country Events

- (1) Provide lunch and water
- (2) Provide appropriate "pit" stops
- (3) Provide a seat in the chase vehicle with an unobstructed view of the solar car
- (4) Provide respect
- (5) Pick up your judge prior to getting your car out of Impound
- (6) Provide your judge with information about what you are going to do *before* you do it!

26.4 A Team's Duty to their Judge on Closed-Track Events

- (1) Provide seat in the chase vehicle in the event that it is necessary to go on to the track.
- (2) Provide respect.
- (3) Pick up your judge prior to getting your car out of the impound.
- (4) Provide your judge with information about what you are going to do before you do it.

27 PENALTIES

Any team failing to comply with the regulations will be penalized. Except for the last day of the race, all penalties will be posted at event headquarters by 8:00 AM the following morning. On the last day of the race, penalties will be posted no later than one hour after the finish of the race.

- 27.1 Disturbing Official Battery Seal** – A battery seal broken without official supervision, in a manner that would allow battery access, will result in the assessment of a 60 mile penalty.
- 27.2 Replacement of Batteries** – Battery replacement will be penalized as provided in Section 18.1 of these regulations.
- 27.3 Non-Solar Charging of Batteries** – Any team using an alternative means to charge their main solar car batteries during the race (other than utilizing solar energy or regenerative brakes) will be disqualified. Teams charging supplemental batteries are required to have a race official present to avoid any misunderstandings, and to prevent the application of this severe penalty.
- 27.4 Failure to Comply with Stops** – Failure to comply with the mandatory stops during each day of the race will incur a 15 mile penalty for each infraction. Teams must drive the two miles preceding and following the Starting Line, the Finish Line, Mandatory Stops, and specified cities. In the event that a vehicle is mechanically unable to drive the two miles, and is in the process of being trailered, no penalty will be assessed *so long as the Team Judge feels that this would be in the best interest of safety*. A team failing to make a required stop forfeits any opportunity to receive that day's Daily Trophy.
- 27.5 Failure to Allow Others to Pass** – Failure to pull over in accordance with Regulation 23 will incur a major penalty, as determined by the appropriate race official.
- 27.6 Traffic Violations** – Any team committing a traffic violation will be penalized. All non-moving traffic violations will result in a 2.5 mile penalty. A moving traffic violation will result in a penalty ranging from 5 miles to disqualification. Any driver who commits two moving violations over the course of the race will be disqualified to drive.

- 27.7 **Failure to Attend a Drivers Meeting** – Failure to attend a driver/advisor meeting each day will incur a 5 mile penalty.
- 27.8 **Conduct** – A race official may assess penalties ranging from 2.5 miles to disqualification for improper or un-sportsman-like conduct. Such conduct may include, but is not limited to, disrespect to race judges and staff, improper language, gestures, questionable attitude, cheating, and use of tobacco products, alcohol, or controlled substances.
- 27.9 **Failure to Comply with Regulations** – A race official may assess penalties ranging from 2.5 miles to disqualification for a team's failure to comply with any regulation, appendix, or *Rules Update* not otherwise specifically set out in these rules.
- 27.10 **Failure to Secure** – Teams failing to secure loose equipment before a convoy begins to move may be penalized up to 5 miles.

28 PROTESTS

Any team desiring to file a protest must do so by submitting a written statement and a \$50 protest fee. The protest must be filed with the Race Director. All protests will be heard by the race jury. The decision of the jury is final and no further appeals will be allowed.

- 28.1 **Time Limit** – Except on the last day, all protests must be filed before 12:00 PM (noon) the day following the incident in question. On the last day of the race, the protest must be filed no later than one hour after the end of the race.
- 28.2 **Opportunity to be Heard** – Representatives from the protesting team will have the opportunity to present their point of view to the race jury at the earliest opportunity.
- 28.3 **Protest Judgments** – If the jury rules in favor of the protesting team, it will notify the Event officials to make the necessary adjustments, and return the protest fee. If the jury rules against the protesting team, it will notify the team of its ruling and take such action with the protest fee as it deems appropriate.
- 28.4 **Jury** – The jury will be composed of the race director and race staff (chosen by the race director).
- 28.5 **Meetings of the Jury** – The jury will meet at the race headquarters each day to consider any questions. It will convene at 7:30 PM, pending the duration of that day's racing. Jury Hearings will be open to the public; jury deliberations will not be open to the public.
- 28.6 **Jury Jurisdiction** – The jury will judge protests on the following: (1) penalties assessed by race officials; (2) conformity of racing with race regulations; and (3) cases which the jury deems appropriate.

- 29 **ADVERTISING/PROMOTION/PUBLICITY** – All advertising/publicity produced by teams or their sponsors will refer to the event as THE WINSTON SOLAR CHALLENGE. By entering the event, all teams and drivers associated with the event agree to the use of their names and pictures in any publicity materials that may be issued by the race organizer.

- 30 **FILM CREWS** – Camera crews or reporters must respect the drivers and their space. These crews need to pay attention to their safety practices in order to prevent a hazardous condition on the road. Alternatively, team sponsors must be aware that the nature of the event attracts publicity, and must take all necessary steps to help prepare their team to handle the demands of the media.

- 31 **APPROPRIATE ACTIONS BY TEAMS** – It is not appropriate for teams, advisors, or sponsors to solicit sponsorship during the race. Teams failing to follow this guideline will be disqualified.

- 32 RESPONSIBILITY TO CHECK RULES UPDATES** – It is the responsibility of every team to acquaint themselves with the specific *Rules Updates* for each race. The race organizer reserves the right to alter, amend, or delete any published rule so long as it gives fair notice in the published *Rules Updates*. Lack of knowledge of new or amended rules is no defense.

Rules Updates will be mailed to each team filing the March 1st Intent to Race Form, and will be published on the WSC Internet site: <http://www.winstonsolar.org>.